

John Albarka Ibrahim

AI Engineer | NLP, RAG and Backend Systems

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SUMMARY

AI engineer with two years of experience building RAG, NLP and backend systems in Python. Built a WhatsApp analytics platform with citation-backed chat and a quantized Nigerian-language sentiment model that runs on CPU. Currently AI/ML Lead at Zibeh Institute of Technology, running applied AI projects and mentoring students.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C/C++

LLM and RAG: RAG, tool calling, agent workflows, embeddings, vector search, prompt engineering, Groq, Google Gemini, OpenAI GPT models, Azure Foundry

ML and NLP: PyTorch, Hugging Face Transformers, TRL, LoRA/QLoRA, scikit-learn, multilingual NLP, ONNX Runtime, INT8 quantization

Backend and Data: FastAPI, Flask, REST APIs, PostgreSQL, pgvector, ChromaDB, FAISS, MySQL, Redis, Redis Streams, Celery, SQLAlchemy, AsyncIO, ETL pipelines, SSE streaming

Tools and Cloud: Docker, Git, GitHub Actions, CI/CD, Azure Container Instances, Langfuse, OpenTelemetry, Hugging Face Hub

Frontend: React, TypeScript, Tailwind CSS

PROJECTS

SentimentScope: WhatsApp Sentiment Analytics and RAG Platform | FastAPI, React, PostgreSQL, Redis, Celery, ChromaDB, ONNX, Docker

- Built a platform that turns WhatsApp chat exports into sentiment analysis and a chat interface that answers questions with source citations.
- Routes each question through four tiers (regex rules, LLM-generated SQL, vector search, LLM synthesis) so simple questions skip the LLM.
- Checks generated SQL with AST validation, a table whitelist and bound parameters. Processes uploads with Celery and Redis and streams answers over SSE.

Joule Dynamics: Agentic Real Estate Market Intelligence | FastAPI, PostgreSQL, pgvector, Groq, Langfuse, OpenTelemetry

- Built a backend that turns plain-English questions into parameterized PostgreSQL and pgvector queries through 18 tools. Tracks 65+ listings in four cities, with data ingested 4 times daily.
- Selects tools per query with local embeddings, cutting tool-schema prompt tokens by over 75%.
- Flags prices that move more than 25% from a 7-day rolling baseline. Traces model calls, token use and latency with Langfuse and OpenTelemetry.
- Runs multi-step tool calls with SSE streaming and model fallback. Tools cover geospatial search, occupancy and pricing analytics, and comparable market analysis (CMA) reports.

AfroXLMR Mini: Nigerian Multilingual Sentiment Model | PyTorch, Hugging Face, ONNX Runtime

- Fine-tuned on 140,256 samples across English, Hausa, Igbo, Yoruba and Nigerian Pidgin, then adapted to 1,058 WhatsApp messages. Scored 94.29% accuracy on a 105-message held-out test set.
- Quantized to INT8 with ONNX Runtime: model size down from 469.9 MB to 112.9 MB, CPU latency down from 42 ms to 8.2 ms per inference.

Smart Image Gallery: Multimodal Product Search | CLIP, FastAPI, HNSW

- Built at Zibeh Institute and integrated into a live e-commerce store with its engineering team. Supports text-to-image and image-to-image search using CLIP embeddings and HNSW, with 85% faster retrieval.
- Reconciles the JSONL persistence layer with the vector index on startup to avoid stale search results.

Nexus Audit Crawler: Distributed Web Crawler | Python, AsyncIO, Redis Streams, PostgreSQL, Playwright

- Built a crawler that coordinates workers with Redis Streams consumer groups and escalates from HTTP requests to Playwright for JavaScript-heavy sites.
- Added per-domain concurrency limits, stale-task reclamation and dead-letter recovery. Covered by 106 automated tests.

Other projects: Google Gemini RAG app for question answering over PDFs, with FAISS retrieval and source citations. Homa (Afrique-Gemma-4B), a team project fine-tuning Gemma-3 4B with QLoRA for an agricultural assistant in four Nigerian languages.

EXPERIENCE

AI/ML Lead, Zibeh Institute of Technology, Jos, Nigeria Jul 2024 - Present

- Lead the institute's AI and data science work, supervising student teams on applied ML, LLM and analytics projects.
- Mentor students from data preparation through model evaluation to final delivery, with guidance on Python, ML workflows and AI APIs.

Virtual AI/ML Trainer, TeKnowledge, Lagos, Nigeria May 2025

- Taught an Azure AI and Microsoft Copilot bootcamp to 50+ participants, who built automation and Q&A prototypes with OpenAI GPT models and Azure services.

AI/ML Trainer, 3MTT with A+ Computer Training Technology Ltd., Bauchi, Nigeria Feb 2024 - May 2024

- Designed and taught a 12-week course on Python, machine learning, computer vision and LLMs to 20+ participants, who built classification, image recognition and chatbot projects.

EDUCATION

B.Eng. Mechanical Engineering, First Class Honors, University of Jos, Nigeria | CGPA 4.63/5.00 2018 - 2023

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (AI-102), Azure AI Fundamentals (AI-900), Azure Data Fundamentals (DP-900), Azure Fundamentals (AZ-900).

COMMUNITY

Microsoft Learn Student Ambassador, 2022 - 2025. Ran workshops on Responsible AI, LLMs and Azure.